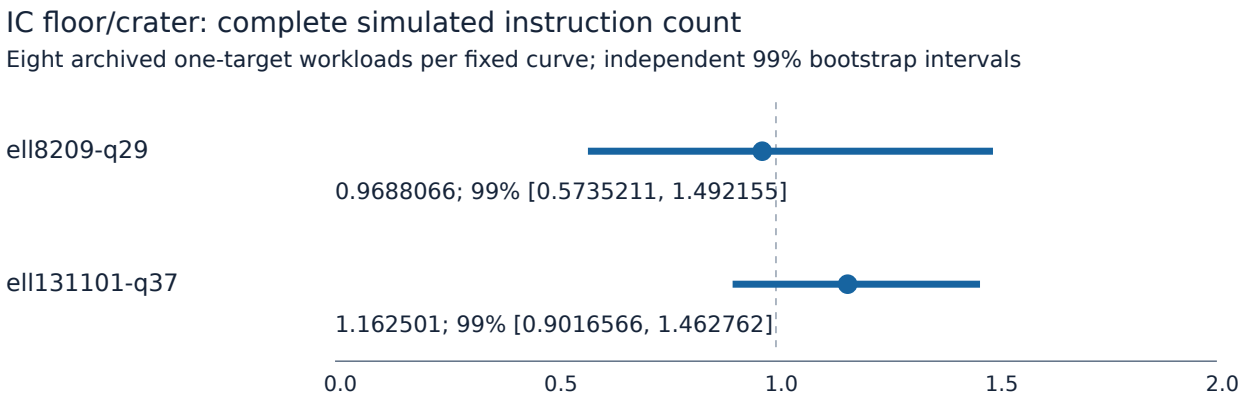
Conductor-gap complete instruction census



Measured on public synthetic prime-field workloads, 2026-10-14 UTC. Classification: accounting and boundary control. Follow-up to PR #1724 (https://github.com/aburan28/crypto/pull/1724).

The existing MITM IC and matched negation-rho implementations are measured in Callgrind simulated user-space instructions over the complete methods::solve call. The interval includes cold setup inside the adapter, queries, relation handling, elimination, recovery and internal verification. It excludes input loading, method resolution, known-answer target construction, process launch, kernel instructions and the external independent point check. These exclusions remain unmeasured; they are not charged zero. No native wall-time speedup is claimed.

Profiles verified against the archived scalars, points and deterministic phase counters: 100/100. The 32 archived workloads remain separate one-target solves. No setup or target transport is shared. The original eight samples per fixed curve are reused; this is accounting on those same points, not a fresh-target replication.

Gap	IC floor/crater Ir	99% interval
ell8209-q29	0.9688066	0.5735211 to 1.492155
ell131101-q37	1.162501	0.9016566 to 1.462762

Frozen repeated instruction-lead criterion: not met. A/A within the frozen 2% gate: true. Maximum absolute relative deviation: 0.00002237726 (fraction). Intervals including one are not equivalence proofs. The estimates are implementation observations on two fixed pairs and cannot establish a conductor effect across isogeny classes.

Comparison status: admissible_implementation_diagnostic. A failed A/A gate makes the instruction comparisons exploratory; failed or missing profiles make inference inadmissible.

ICV1 model	IC / matched rho Ir	Paired 99% interval
icv1-fp29-t47-14392892	28.15254	14.01229 to 74.39444
icv1-fp29-t47-1f934956	25.27633	13.30118 to 48.24704
icv1-fp37-t277-91fd6e29	47.77148	34.84198 to 79.65082
icv1-fp37-t277-090dc414	47.07830	29.44940 to 104.8381

These ratios compare the same one-target workloads within the frozen instruction interval; excluded work and native wall time remain unmeasured.

See native per-curve and per-target tables (analysis.md), machine-readable analysis (analysis.json), frozen protocol (PROTOCOL.md), amended protocol (PROTOCOL-v2.md), frozen jobs and original records (jobs.json), lossless raw evidence (raw-callgrind.tar.xz), execution commands (COMMANDS.md), validation (VALIDATION.md) and PDF (report.pdf). The prior study's exact EC1/UID representation manifests and Sage evidence remain at the archived input study (../ecbench_conductor_gap_20261012/README.md). No map is constructed and no transfer cost is assumed.

Run reports

- ell8209-q29 run report (reports/ell8209-q29/README.md), PDF (reports/ell8209-q29/report.pdf).

- ell131101-q37 run report (reports/ell131101-q37/README.md), PDF (reports/ell131101-q37/report.pdf).

Requirements and unresolved work

Requirement	Status and evidence
Continue the large-gap experiment	Whole-solve instruction census on all archived one-target workloads
Previously unpriced user-space implementation work	Included in the Ir interval; kernel/hardware work remains outside it
Scalar correctness and deterministic phase replay	Native validation in analysis.json; another-host replay remains open
Prime-field F4/SAT comparison	Not implemented in the native prime pipeline; remains open
Explicit isogeny and transport costs	Not attempted; remains open
Representation controls and more independent classes	Not attempted; two fixed pairs cannot establish a structural effect
Intrinsic or cryptographic-scale hardness	Unresolved; this census supplies no general security conclusion

Interrupted attempts: launch before preregistration (attempt-unpreregistered/README.md) and phase-dump volume obstruction (attempt-phase-dumps/README.md). Both retain all raw files and supply no primary estimate. The profiling obstruction (../../problems/conductor-gap-callgrind-phase-dump-volume.md) and missing prime-field algebraic oracle (../../problems/prime-field-conductor-gap-algebraic-oracle.md) track their exact obligations.

Canonical visual scope

The scoreboard receives a separate instruction-unit panel. The earlier GAE panel and figures remain historical. No new curve, map or conductor is constructed, so conductor/isogeny graph topology is unchanged. No generic GAE floor, scaling exponent, native wall-time leaderboard, or hypothesis state is promoted.